CSE 4096 Introduction to Machine Learning
Projects

Project Content: A project usually consists of comparing performances of different algorithms, and perhaps proposing your own algorithms/improvements, and implementing them.

Project Team: Each project will be performed by a group of at most two students.

Project Stages: The stages for the completion of the project are as follows:

1. Choose a project topic and write a project plan (**due April 3, 2023**)
2. Conduct the project and write a mid-term report (**due May 8, 2023)**
3. Write a final project report (**due June 5, 2023**)
4. Present the project in class (**due June 5 and June 8, 2023**)

Preparation of the Project Reports: Please provide:

1. A typed report describing what you did that includes figures, graphs, images, references and commented source code.
2. The style of the report should follow the style of an IEEE journal article with single column format. You must use the IEEE style for citing and listing the references.
3. Please include commented copies of your program **source code in an Appendix** as part of your report.
4. You can submit a printed copy your report during class hours **(optional)**.
5. You should submit your final report (as well as the project plan and midterm report) via the classroom web page. Please **zip all of your files (report + code) into a single file** before submitting. The zip file should have a name like: "CSE4088_Project_Surname.zip".
6. For your source code please include a "readme" file describing the necessary steps for using your program properly.
7. Late reports will be penalized.

Project plan: The project plan should include the following:

1. Project title and team members
2. **An abstract:** in one paragraph or two, describe briefly the project scope and what you hope to accomplish.
3. **Project schedule:** It should include a list of subtasks that you plan to accomplish, for each subtask, describes how do you plan to approach it, and the deadline for completing it. You should allocate time for searching the project subject, initial literature search, preparing the midterm report, final report, and final presentation, in addition to the reading/simulation work you plan to do.
4. **List of references:** this includes papers and documents and possibly web links that you have found that are relevant to your project topic. Please include complete citation of each item: author, title of the article, journal or book name, and if appropriate the URL.

Project Midterm Report: Describe the accomplishment so far and remaining work to be completed in 3-5 pages. You should think of the mid-term project report as an on-going document that starts from your project plan and will evolve into the final report. The midterm report should include the following:

1. Project title and team members
2. **An abstract** (this should be a revised version of what you have written in your project plan).
3. **What you have accomplished so far:** This can be divided into multiple sections, each section describing one subtask that you have worked on. For each subtask, describing what you have done so far, what you have learnt.

4. **Remaining work:** This section should outline the remaining work to be done in each subtask and the revised schedule for them. (if different from the project plan).
5. **Revised Project plan:** you may want to revise the original plan based on your progress so far. You may need to modify the deadlines only, or you may even need to modify the sub-tasks. The later may be true if after your study so far, you want to redirect your project in terms of the technical content.
6. **List of references:** you may add new references and delete irrelevant ones that originally appeared in the project plan.

Final project report: The final report is an extension of your midterm report and should include the following in (5-10 pages)

1. **Project title** and team member(s)
2. An **abstract**
3. **Overview** of the project and schedule (describe the subtasks and completion schedule. Including who is responsible for which subtask if you have multiple members on the team.)
4. **Project accomplishment:** This can be divided into multiple sections, each section describing one subtask that you have worked on. For each subtask, describing what you have done, what you have learnt. If your project includes the comparison of different systems/techniques, you may have separate sections describing each system/technique and another section comparing these systems/techniques and summarizes the pros and cons of each. You may have another section describing your proposed solution on how to modify each or combine them.
5. **Summary:** This section should briefly summarize your findings. It should also include a discussion of future works that you think should be studied.
6. **List of references:** update from your midterm report as appropriate.

Final project presentation: A 10-15 minute oral presentation, possibly with demonstration of computer implementations. Time allocation depends on the number of projects to be presented and the number of team members.

You should prepare a presentation file using a computer software (e.g. powerpoint).

1. You will be giving the presentation in class.
2. If you want to include any demonstration in your presentation, please make sure it can be presented directly from the computer.
3. The suggested outline for your presentation:
    a. Project title and team member(s)
    b. Overview of your project: the overall goal and subtasks, and who is responsible for what. (The finalized schedule)
    c. Describe each subtask: what you did and what you learnt/designed.
    d. Demos if appropriate
    e. Summary

Possible Journals/Conferences/Resources that might be helpful for topic selection

Journals and Conferences:

1. IEEE Transactions on Pattern Analysis and Machine Intelligence
2. Pattern Recognition
3. Pattern Recognition Letters
4. International Journal of Pattern Recognition and Artificial Intelligence
5. International Conference on Pattern Recognition (ICPR)
6. Neural Information Processing Systems (NEURIPS)
7. International Conference on Learning Representations (ICLR)
8. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR) papers
    a. http://openaccess.thecvf.com

Datasets:

9. UCI Machine Learning Repository : https://archive.ics.uci.edu/ml/index.php
10. Kaggle challenges: A machine learning competition website: https://www.kaggle.com/competitions